

```
In [1]: import networkx as nx
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import math
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

```
In [4]: df = pd.read_csv("data2.csv")
print(df.head())
```

	Place_Name	Latitude	Longitude
0	Graphic Era Deemed to be University, Dehradun	30.2685	77.9968
1	ISBT Dehradun	30.2861	77.9954
2	Clement Town, Dehradun	30.2662	78.0075
3	Sahranpur Road, Dehradun	30.2988	77.9945
4	Clock Tower, Dehradun	30.3244	78.0411

```
In [5]: df
```

```
Out[5]:
```

	Place_Name	Latitude	Longitude
0	Graphic Era Deemed to be University, Dehradun	30.2685	77.9968
1	ISBT Dehradun	30.2861	77.9954
2	Clement Town, Dehradun	30.2662	78.0075
3	Sahranpur Road, Dehradun	30.2988	77.9945
4	Clock Tower, Dehradun	30.3244	78.0411
...	...	...	...
2362	Bageshwar	29.8415	79.7712
2363	Kanjhawala Crossing	28.7211	NaN
2364	Kashipur	29.2104	78.9618
2365	NaN	25.0451	79.4451
2366	Bamrauli Airport	25.4411	NaN

2367 rows × 3 columns

```
In [6]: df.shape
```

```
Out[6]: (2367, 3)
```

```
In [7]: df.size
```

```
Out[7]: 7101
```

```
In [8]: df.columns
```

```
Out[8]: Index(['Place_Name', 'Latitude', 'Longitude'], dtype='object')
```

```
In [9]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2367 entries, 0 to 2366
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Place_Name  2244 non-null   object
1   Latitude    2254 non-null   float64
2   Longitude   2259 non-null   float64
dtypes: float64(2), object(1)
memory usage: 55.6+ KB
```

```
In [10]: df.describe()
```

Out[10]:

		Latitude	Longitude
	count	2254.000000	2259.000000
	mean	27.691830	79.804069
	std	3.091533	4.241633
	min	20.701100	8.000000
	25%	25.241100	77.201100
	50%	28.583100	78.041100
	75%	30.321100	82.631100
	max	34.551100	95.351100

```
In [11]: print("Duplicate rows:", df.duplicated().sum())
```

Duplicate rows: 15

```
In [12]: # Remove duplicates
df = df.drop_duplicates()
df
```

Out[12]:

	Place_Name	Latitude	Longitude
0	Graphic Era Deemed to be University, Dehradun	30.2685	77.9968
1	ISBT Dehradun	30.2861	77.9954
2	Clement Town, Dehradun	30.2662	78.0075
3	Sahranpur Road, Dehradun	30.2988	77.9945
4	Clock Tower, Dehradun	30.3244	78.0411
...	...	...	...
2353	Kanhaiya Nagar	28.6811	77.1511
2358	Banjarawala	NaN	78.0311
2360	NaN	26.9911	75.7611
2363	Kanjhawala Crossing	28.7211	NaN
2366	Bamrauli Airport	25.4411	NaN

2352 rows × 3 columns

```
In [13]: print("Duplicate rows:", df.duplicated().sum())
```

Duplicate rows: 0

```
In [14]: print(df.isnull().sum())
```

```
Place_Name    120
Latitude      113
Longitude     108
dtype: int64
```

```
In [15]: # Remove null values
df = df.dropna()
```

```
In [16]: print(df.isnull().sum())
```

```
Place_Name    0
Latitude      0
Longitude      0
dtype: int64
```

```
In [17]: df.loc[:, "Place_Name"] = df["Place_Name"].fillna("UNKNOWN")
df
```

Out[17]:

	Place_Name	Latitude	Longitude
0	Graphic Era Deemed to be University, Dehradun	30.2685	77.9968
1	ISBT Dehradun	30.2861	77.9954
2	Clement Town, Dehradun	30.2662	78.0075
3	Sahranpur Road, Dehradun	30.2988	77.9945
4	Clock Tower, Dehradun	30.3244	78.0411
...	...	...	...
2343	Bandgaon Crossing	22.8811	85.3811
2344	Chakradharpur RS North	22.7011	85.6211
2345	Chaibasa Bypass North	22.5611	85.8111
2346	Jhinkpani RS North	22.4111	8.0000
2353	Kanhaiya Nagar	28.6811	77.1511

2030 rows × 3 columns

```
In [18]: df.loc[:, "Latitude"] = df["Latitude"].fillna(df["Latitude"].mean())
df.loc[:, "Longitude"] = df["Longitude"].fillna(df["Longitude"].mean())
df
```

Out[18]:

	Place_Name	Latitude	Longitude
0	Graphic Era Deemed to be University, Dehradun	30.2685	77.9968
1	ISBT Dehradun	30.2861	77.9954
2	Clement Town, Dehradun	30.2662	78.0075
3	Sahranpur Road, Dehradun	30.2988	77.9945
4	Clock Tower, Dehradun	30.3244	78.0411
...	...	...	...
2343	Bandgaon Crossing	22.8811	85.3811
2344	Chakradharpur RS North	22.7011	85.6211
2345	Chaibasa Bypass North	22.5611	85.8111
2346	Jhinkpani RS North	22.4111	8.0000
2353	Kanhaiya Nagar	28.6811	77.1511

2030 rows × 3 columns

In [19]: `df.shape`

Out[19]: (2030, 3)

In [20]: `# valid coordinate
df = df[(df['Latitude'].between(-90, 90)) & (df['Longitude'].between(-180, 180))]`

In [21]: `df['Latitude'] = df['Latitude'].astype(float)
df['Longitude'] = df['Longitude'].astype(float)
df`

Out[21]:

	Place_Name	Latitude	Longitude
0	Graphic Era Deemed to be University, Dehradun	30.2685	77.9968
1	ISBT Dehradun	30.2861	77.9954
2	Clement Town, Dehradun	30.2662	78.0075
3	Sahranpur Road, Dehradun	30.2988	77.9945
4	Clock Tower, Dehradun	30.3244	78.0411
...	...	...	...
2343	Bandgaon Crossing	22.8811	85.3811
2344	Chakradharpur RS North	22.7011	85.6211
2345	Chaibasa Bypass North	22.5611	85.8111
2346	Jhinkpani RS North	22.4111	8.0000
2353	Kanhaiya Nagar	28.6811	77.1511

2030 rows × 3 columns

In [22]: `df["Place_Name"] = df["Place_Name"].str.upper().str.strip()`

In [23]: `z1 = (df["Latitude"] - df["Latitude"].mean()) / df["Latitude"].std()
z2 = (df["Longitude"] - df["Longitude"].mean()) / df["Longitude"].std()
df = df[(np.abs(z1) <= 3) & (np.abs(z2) <= 3)]
df`

Out[23]:

	Place_Name	Latitude	Longitude
0	GRAPHIC ERA DEEMED TO BE UNIVERSITY, DEHRADUN	30.2685	77.9968
1	ISBT DEHRADUN	30.2861	77.9954
2	CLEMENT TOWN, DEHRADUN	30.2662	78.0075
3	SAHRANPUR ROAD, DEHRADUN	30.2988	77.9945
4	CLOCK TOWER, DEHRADUN	30.3244	78.0411
...	...	...	...
2342	KHUNTI MARKET NORTH	23.0711	85.2811
2343	BANDGAON CROSSING	22.8811	85.3811
2344	CHAKRADHARPUR RS NORTH	22.7011	85.6211
2345	CHAIBASA BYPASS NORTH	22.5611	85.8111
2353	KANHAIYA NAGAR	28.6811	77.1511

2019 rows × 3 columns

In [24]: `df.shape`

Out[24]: (2019, 3)

In [25]:

```

Q1_lat = df["Latitude"].quantile(0.25)
Q3_lat = df["Latitude"].quantile(0.75)
IQR_lat = Q3_lat - Q1_lat
df = df[(df["Latitude"] >= Q1_lat - 1.5*IQR_lat) &
(df["Latitude"] <= Q3_lat + 1.5*IQR_lat)]
df

```

Out[25]:

	Place_Name	Latitude	Longitude
0	GRAPHIC ERA DEEMED TO BE UNIVERSITY, DEHRADUN	30.2685	77.9968
1	ISBT DEHRADUN	30.2861	77.9954
2	CLEMENT TOWN, DEHRADUN	30.2662	78.0075
3	SAHRANPUR ROAD, DEHRADUN	30.2988	77.9945
4	CLOCK TOWER, DEHRADUN	30.3244	78.0411
...	...	...	...
2342	KHUNTI MARKET NORTH	23.0711	85.2811
2343	BANDGAON CROSSING	22.8811	85.3811
2344	CHAKRADHARPUR RS NORTH	22.7011	85.6211
2345	CHAIBASA BYPASS NORTH	22.5611	85.8111
2353	KANHAIYA NAGAR	28.6811	77.1511

2019 rows × 3 columns

In [26]:

```

# Min-Max Scaling
minmax = MinMaxScaler()

```

```
df_minmax = df.copy()
df_minmax[["Latitude", "Longitude"]] = minmax.fit_transform(df[["Latitude", "Longitude"]])
df_minmax
```

Out[26]:

	Place_Name	Latitude	Longitude
0	GRAPHIC ERA DEEMED TO BE UNIVERSITY, DEHRADUN	0.690787	0.235867
1	ISBT DEHRADUN	0.692058	0.235790
2	CLEMENT TOWN, DEHRADUN	0.690621	0.236456
3	SAHRANPUR ROAD, DEHRADUN	0.692975	0.235740
4	CLOCK TOWER, DEHRADUN	0.694823	0.238305
...	...	...	...
2342	KHUNTI MARKET NORTH	0.171119	0.636764
2343	BANDGAON CROSSING	0.157401	0.642267
2344	CHAKRADHARPUR RS NORTH	0.144404	0.655476
2345	CHAIBASA BYPASS NORTH	0.134296	0.665933
2353	KANHAIYA NAGAR	0.576173	0.189323

2019 rows × 3 columns

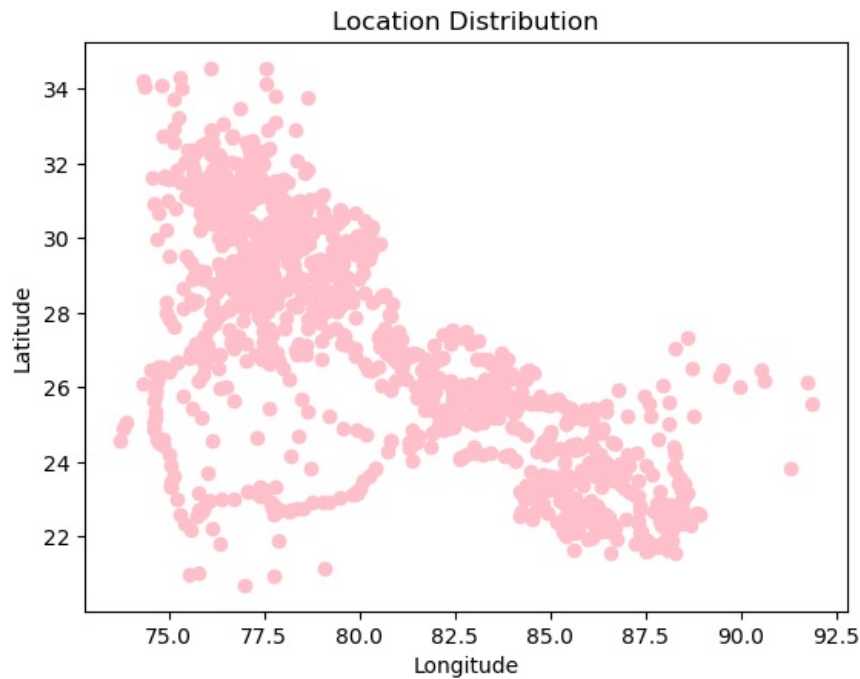
```
In [27]: # Standard Scaling
std = StandardScaler()
df_standard = df.copy()
df_standard[["Latitude", "Longitude"]] = std.fit_transform(df[["Latitude", "Longitude"]])
df_standard
```

Out[27]:

	Place_Name	Latitude	Longitude
0	GRAPHIC ERA DEEMED TO BE UNIVERSITY, DEHRADUN	0.828070	-0.462064
1	ISBT DEHRADUN	0.833752	-0.462425
2	CLEMENT TOWN, DEHRADUN	0.827327	-0.459303
3	SAHRANPUR ROAD, DEHRADUN	0.837852	-0.462657
4	CLOCK TOWER, DEHRADUN	0.846118	-0.450633
...	...	...	...
2342	KHUNTI MARKET NORTH	-1.495647	1.417487
2343	BANDGAON CROSSING	-1.556990	1.443290
2344	CHAKRADHARPUR RS NORTH	-1.615104	1.505217
2345	CHAIBASA BYPASS NORTH	-1.660304	1.554242
2353	KANHAIYA NAGAR	0.315570	-0.680278

2019 rows × 3 columns

```
In [28]: plt.scatter(df["Longitude"], df["Latitude"], color="pink")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.title("Location Distribution")
plt.show()
```



```
In [29]: df.shape
```

```
Out[29]: (2019, 3)
```

```
In [30]: df.size
```

```
Out[30]: 6057
```

```
In [31]: corr = df[["Latitude", "Longitude"]].corr()
print(corr)
```

	Latitude	Longitude
Latitude	1.000000	-0.693517
Longitude	-0.693517	1.000000

```
In [32]: df = df.copy()
df = df.dropna(subset=["Place_Name", "Latitude", "Longitude"])
df = df.reset_index(drop=True)
```

```
In [33]: coords = dict(zip(df["Place_Name"], zip(df["Latitude"], df["Longitude"])))
```

```
In [ ]: # K-MEANS CLUSTERING
kmeans = KMeans(n_clusters=3)
df["Cluster"] = kmeans.fit_predict(df[["Latitude", "Longitude"]])
```

```
In [ ]: print("\nCluster Assignment:")
print(df[["Place_Name", "Cluster"]].head())
```

```
In [34]: # Step 1: Build graph
graph = {}

for i in range(len(df)):
    city1 = df.loc[i, "Place_Name"]
    lat1 = df.loc[i, "Latitude"]
    lon1 = df.loc[i, "Longitude"]

    distances = []

    for j in range(len(df)):
        if i == j:
```

```

        continue

    city2 = df.loc[j, "Place_Name"]
    lat2 = df.loc[j, "Latitude"]
    lon2 = df.loc[j, "Longitude"]

    dist = ((lat2 - lat1)**2 + (lon2 - lon1)**2)**0.5

    # CLUSTERING
    if df.iloc[i]["Cluster"] != df.iloc[j]["Cluster"]:
        dist += 1

    distances.append((city2, dist))

distances.sort(key=lambda x: x[1])
graph[city1] = [city for city, _ in distances[:3]]

# Step 2: MAKE GRAPH BIDIRECTIONAL ✓
for city in graph:
    for neighbor in graph[city]:
        if city not in graph[neighbor]:
            graph[neighbor].append(city)

```

In [35]: `print(list(graph.keys())[:20])`

```

['GRAPHIC ERA DEEMED TO BE UNIVERSITY, DEHRADUN', 'ISBT DEHRADUN', 'CLEMENT TOWN, DEHRADUN', 'SAHRANPUR ROAD, DEHRADUN', 'CLOCK TOWER, DEHRADUN', 'RAJPUR ROAD, DEHRADUN', 'BALLUPUR CHOWK, DEHRADUN', 'PREM NAGAR, DEHRADUN', 'MUSSOORIE LIBRARY CHOWK', 'LAL TIBBA, MUSSOORIE', 'SELAKUI INDUSTRIAL AREA', 'HERBERTPUR', 'PAONTA SAHIB', 'CHHUTMALPUR', 'SAHARANPUR JUNCTION', 'ROORKEE IIT CAMPUS', 'ROORKEE CIVIL LINES', 'HARIDWAR HAR KI PAURI', 'PATANJALI YOGPEETH, HARIDWAR', 'RISHIKESH LAXMAN JHULA']

```

In [36]: `print("Total cities:", len(graph))`
`print("Sample:", list(graph.items())[:3])`

```

Total cities: 2019
Sample: [('GRAPHIC ERA DEEMED TO BE UNIVERSITY, DEHRADUN', ['SUBHASH NAGAR, DEHRADUN', 'SMITH NAGAR', 'SUBHASH NAGAR MARKET', 'TURNER ROAD', 'GRAPHIC ERA HOSPITAL']), ('ISBT DEHRADUN', ['LALPUL DEHRADUN', 'SAHRANPUR ROAD, DEHRADUN', 'MAJRA, DEHRADUN']), ('CLEMENT TOWN, DEHRADUN', ['KANDOLI', 'SOCIETY AREA CLEMENT TWN', 'SGRR CLEMENT TOWN', 'MINDROLLING MONASTERY', 'CLEMENT TOWN MARKET', 'LAKSHYA CLASSES CLEMENT'])]

```

In [37]: `def haversine(lat1, lon1, lat2, lon2):`
`R = 6371 # km`

```

    lat1, lon1, lat2, lon2 = map(math.radians, [lat1, lon1, lat2, lon2])

    dlat = lat2 - lat1
    dlon = lon2 - lon1

    a = math.sin(dlat/2)**2 + math.cos(lat1)*math.cos(lat2)*math.sin(dlon/2)**2
    c = 2 * math.asin(math.sqrt(a))

    return R * c

```

In [38]: `def heuristic(lat1, lon1, lat2, lon2):`
`return ((lat2 - lat1)**2 + (lon2 - lon1)**2)**0.5 * 111`

In [39]: `import heapq`

```

def astar(graph, start, goal, coords):
    open_list = []
    heapq.heappush(open_list, (0, start))

    g_cost = {node: float('inf') for node in graph}
    g_cost[start] = 0

    parent = {}

    while open_list:
        _, current = heapq.heappop(open_list)

        # Goal reached
        if current == goal:

```



```

        path = []
        while current in parent:
            path.append(current)
            current = parent[current]
        path.append(start)
        return path[::-1], g_cost[goal]

    for neighbor in graph[current]:
        lat1, lon1 = coords[current]
        lat2, lon2 = coords[neighbor]

        # g(n)
        cost = g_cost[current] + haversine(lat1, lon1, lat2, lon2)

        if cost < g_cost[neighbor]:
            g_cost[neighbor] = cost

            # h(n)
            latg, longg = coords[goal]
            h = heuristic(lat2, lon2, latg, longg)

            # f(n)
            f = cost + h

            heapq.heappush(open_list, (f, neighbor))
            parent[neighbor] = current

    return None, float('inf')

```

```

In [40]: start = input("Enter start city: ").strip().upper()
goal = input("Enter goal city: ").strip().upper()

if start not in graph or goal not in graph:
    print("Invalid city name!")
else:
    path, distance = astar(graph, start, goal, coords)

    print("Shortest Path:", " → ".join(path))
    print("Total Distance:", round(distance, 2), "km")

```

Shortest Path: SMITH NAGAR → SUBHASH NAGAR, DEHRADUN → GRAPHIC ERA HOSPITAL → TURNER ROAD → MAJRA, DEHRADUN → LALPUL DEHRADUN → SAHRANPUR ROAD, DEHRADUN → NIRANJANPUR → KAONLI ROAD → KANWALI → GMS ROAD → INDRESH HOSPITAL → SARSARI → BANJARAWALA → BHARUWALA → KANDOLI → CLEMENT TOWN, DEHRADUN → CLEMENT TOWN MARKET
Total Distance: 16.96 km

```

In [41]: path, distance = astar(graph, start, goal, coords)

# Fix nested issue (safety)
path = [p if isinstance(p, str) else p[0] for p in path]

print("Shortest Path:", " → ".join(path))
print("Total Distance:", round(distance, 2), "km")

```

Shortest Path: SMITH NAGAR → SUBHASH NAGAR, DEHRADUN → GRAPHIC ERA HOSPITAL → TURNER ROAD → MAJRA, DEHRADUN → LALPUL DEHRADUN → SAHRANPUR ROAD, DEHRADUN → NIRANJANPUR → KAONLI ROAD → KANWALI → GMS ROAD → INDRESH HOSPITAL → SARSARI → BANJARAWALA → BHARUWALA → KANDOLI → CLEMENT TOWN, DEHRADUN → CLEMENT TOWN MARKET
Total Distance: 16.96 km

```

In [46]: features = []

for i in range(len(df)):
    for j in range(len(df)):
        if i == j:
            continue

        lat1, lon1 = df.loc[i, ["Latitude", "Longitude"]]
        lat2, lon2 = df.loc[j, ["Latitude", "Longitude"]]

        dist = math.sqrt((lat1 - lat2)**2 + (lon1 - lon2)**2)

        # Derived features
        traffic = np.random.uniform(0, 1)
        road_quality = np.random.uniform(0, 1)

```

```
# Label (classification target)
label = 1 if (traffic < 0.5 and road_quality > 0.5) else 0

features.append([dist, traffic, road_quality, label])

clf_df = pd.DataFrame(features, columns=["distance", "traffic", "road_quality", "label"])
```

```
In [47]: from sklearn.linear_model import LogisticRegression

X = clf_df[["distance", "traffic", "road_quality"]]
y = clf_df["label"]

model = LogisticRegression()
model.fit(X, y)
```

```
Out[47]: ▼ LogisticRegression ⓘ ?
          ► Parameters
```

```
In [48]: print("Slope (coefficient):", model.coef_)
          print("Intercept:", model.intercept_)

Slope (coefficient): [[-5.05225714e-04 -1.24299892e+01  1.24329512e+01]]
Intercept: [-3.82331849]
```

```
In [51]: print("R2 Score:", model.score(X, y))

R2 Score: 0.915657055789622
```

```
In [52]: y_pred = model.predict(X)

for i in range(len(y)):
    print("Actual:", y[i], "Predicted:", y_pred[i])
```

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
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Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

... [110601 lines truncated] ...

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 1
Actual: 0 Predicted: 0
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Actual: 1 Predicted: 1
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Actual: 1 Predicted: 1
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Actual: 0 Predicted: 0
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Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

... [119452 lines truncated] ...

```
Actual: 0 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted:
```

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`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
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Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

... [118798 lines truncated] ...

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

... [118746 lines truncated] ...

```
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual:
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
... [119254 lines truncated] ...
```

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 1
Actual: 0
```

IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

```
Current values:
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
ServerApp.rate_limit_window=3.0 (secs)
```

```
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
```

... [118794 lines truncated] ...

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

[illegible]

```
Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0
```

```
IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub_data_rate_limit`.

Current values:
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
ServerApp.rate_limit_window=3.0 (secs)
```

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

... [117712 lines truncated] ...

```
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 1
Actual: 1 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

... [118110 lines truncated] ...

```
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

[illegible][illegible]

IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

```
Current values:
ServerApp.iopub_data_rate_limit=10000000.0 (bytes/sec)
ServerApp.rate_limit_window=3.0 (secs)
```

```
Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
```

... [117454 lines truncated] ...

```
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
```

IOPub data rate exceeded.

The Jupyter server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable

`--ServerApp.iopub\_data\_rate\_limit`.

Current values:

ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

ServerApp.rate_limit_window=3.0 (secs)

```
0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1
```

... [116543 lines truncated] ...

```
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual:
```

IOPub data rate exceeded.
The Jupyter server will temporarily stop sending output
to the client in order to avoid crashing it.
To change this limit, set the config variable
`--ServerApp.iopub\_data\_rate\_limit`.

Current values:
ServerApp.iopub_data_rate_limit=1000000.0 (bytes/sec)
ServerApp.rate_limit_window=3.0 (secs)

```
In [55]: print(model.coef_)
         print(model.score(X, y))
```

```
[[-5.05225714e-04 -1.24299892e+01  1.24329512e+01]]
0.915657055789622
```

```
In [ ]:
```